
Full Title: Single-Word Speech Intelligibility for Children with Repaired Cleft Lip and Palate
Abbreviated Title: Single-Word Speech Intelligibility
Keywords: Speech intelligibility, cleft lip
Lead/Corresponding Author: David J. Zajac
Manuscript Affiliation: Department of Craniofacial and Surgical Care
Statisticians: Dan Huang, Sophie Shan, Wanbing Wang, Esther Zhou

AI USE DISCLAIMER

Portions of this statistical analysis plan were drafted with the assistance of Claude (Anthropic), a large language model AI assistant. All AI-generated content was reviewed, verified, and approved by the study statisticians. The AI was used solely as a writing and organizational aid; all statistical decisions, methodology, and scientific judgment remain the sole responsibility of the authors.

Contents

AIMS	2
STUDY DESIGN SUMMARY	2
KEY REFERENCES	2
TABLES AND FIGURES SUMMARY	3
ANALYTIC FILE	3
Inclusions/Exclusions	3
Definitions: Outcomes, Exposures, Covariates	3
PRELIMINARY ANALYSIS	6
Key decisions	6
PRIMARY ANALYSIS	6
Key decisions	6
Missing Data	6
Significance level/Multiple Comparisons	7
Sensitivity Analysis	7
SECONDARY ANALYSIS	7
Key decisions	7
Significance level/Multiple Comparisons	8

AIMS

Aim 1: Assess differences in overall speech intelligibility between children with CLP, and children in the CP, OM, and TD groups.

Aim 2: Evaluate whether patterns of phonetic errors differ between children with CLP and children in the CP, OM, and TD groups.

STUDY DESIGN SUMMARY

Single-center, cross-sectional observational study including 115 children from four groups: children with repaired cleft lip and palate (CLP), repaired cleft palate only (CP), a history of otitis media (OM), and typically developing (TD) children. Speech intelligibility was assessed using a 154-word single-word intelligibility test in which adult listeners independently transcribed recorded speech samples. Listeners were organized into groups of three, with each group evaluating recordings from approximately 5–6 children. Phonetic errors were recorded based on listener transcriptions and categorized into 11 predefined phonetic contrasts.

KEY REFERENCES

1. Zajac DJ, Plante C, Lloyd A, Haley KL. Reliability and validity of a computer-mediated, single-word intelligibility test: preliminary findings for children with repaired cleft lip and palate. *Cleft Palate Craniofac J.* 2011;48(5):538–549. doi:10.1597/09-166
2. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med.* 2016;15(2):155–163. doi:10.1016/j.jcm.2016.02.012
3. Bretz F, Maurer W, Brannath W, Posch M. A graphical approach to sequentially rejective multiple test procedures. *Stat Med.* 2009;28(4):586–604. doi:10.1002/sim.3495
4. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc B.* 1995;57(1):289–300.

TABLES AND FIGURES SUMMARY

MANUSCRIPT

Table/Figure #	Brief title	JOB	Notes or comments
Table 1	Baseline characteristics and mean speech intelligibility by group	Descriptive summary table	Includes N, age (mean $\pm$ SD), sex (n, %), and mean intelligibility score (averaged across L1–L3) by group (CLP, CP, OM, TD); no statistical testing
Table 2	Adjusted association between cleft type and speech intelligibility	Inferential analysis	LMM of mean intelligibility on cleft type, adjusted for age and sex; CLP as reference group
Table 3	Association Between Group and Contrast-Specific Intelligibility Scores	Assess whether intelligibility differs by groups for each phonetic contrast.	Linear regression adjusted for age and sex; estimates shown with 95% CI
Figure 1	Distribution of Mean Speech Intelligibility Scores by Group and Sex	Compare intelligibility across groups and sex.	Boxplots show median, IQR, and range by group and sex.
Figure 2	Mean Speech Intelligibility vs. Age by Group	Assess the relationship between age and intelligibility across groups.	Scatter plot with points colored by group, points represent individuals. Optional trend lines may be added.
Suppl. Table 1	Inter-Rater Reliability of Listener Groups (Intra-class Correlation Coefficients)	Assess agreement among listeners.	ICC quantifies inter-rater reliability; higher values indicate stronger agreement.
Suppl. Table 2	Sensitivity analyses for cohort effect on mean speech intelligibility	Assess robustness of primary model results	Compares estimates from the primary LMM with sensitivity analyses including simplified random effects, VP exclusion, and OLS model to assess robustness.

ANALYTIC FILE

Inclusions/Exclusions

Full cohort (N=115) is retained for analysis. No participants are excluded.

Data note: Responses from Listener 74 for participant 30310 were discarded prior to data receipt due to audio playback problems with the transcription program. As a result, M_SWI for participant 30310 reflects the mean of two listeners rather than three. This participant is retained in all analyses.

Definitions: Outcomes, Exposures, Covariates

OUTCOMES

Outcome (friendly name)	Values and Format	Definition or variable name in dataset	Notes or QC comments
-------------------------	-------------------	--	----------------------

DERIVED VARIABLES IN THIS MANUSCRIPT

M_SWI	0–100 (%); continuous	Mean of L1_SWI, L2_SWI, and L3_SWI; percentage of 154 test words correctly transcribed, averaged across 3 listeners	Main outcome; column M
alv_pal	0–100 (%); continuous	Mean % correct for alveolar-palatal sibilant contrast (14 words; e.g., <i>see-she</i>), averaged across 3 listeners	Secondary outcome; column R
alv_velar	0–100 (%); continuous	Mean % correct for alveolar-velar stop contrast (14 words; e.g., <i>tap-cap</i>), averaged across 3 listeners	Secondary outcome; column S
cl_sing	0–100 (%); continuous	Mean % correct for cluster-singleton contrast (14 words; e.g., <i>stop-top</i>), averaged across 3 listeners	Secondary outcome; column T
con_null	0–100 (%); continuous	Mean % correct for initial consonant-null contrast (14 words; e.g., <i>pat-at</i>), averaged across 3 listeners	Secondary outcome; column U
fr_bk	0–100 (%); continuous	Mean % correct for front-back vowel contrast (14 words; e.g., <i>fill-full</i>), averaged across 3 listeners	Secondary outcome; column V
fric_affr	0–100 (%); continuous	Mean % correct for fricative-affricate contrast (14 words; e.g., <i>ship-chip</i>), averaged across 3 listeners	Secondary outcome; column W
hi_lo	0–100 (%); continuous	Mean % correct for high-low vowel contrast (14 words; e.g., <i>beat-bit</i>), averaged across 3 listeners	Secondary outcome; column X
r_w	0–100 (%); continuous	Mean % correct for liquid-glide contrast (14 words; e.g., <i>ray-way</i>), averaged across 3 listeners	Secondary outcome; column Y
st_fric-affr	0–100 (%); continuous	Mean % correct for stop-fricative/affricate contrast (14 words; e.g., <i>base-vase</i>), averaged across 3 listeners	Secondary outcome; column Z
st_nas	0–100 (%); continuous	Mean % correct for stop-nasal contrast (14 words; e.g., <i>bop-mop</i>), averaged across 3 listeners	Secondary outcome; column AA
vd_vl	0–100 (%); continuous	Mean % correct for voiced-voiceless stop contrast (14 words; e.g., <i>big-pig</i>), averaged across 3 listeners	Secondary outcome; column AB

EXISTING DERIVED VARIABLES

L1_SWI	0–100 (%); continuous	Percentage of 154 test words correctly transcribed by Listener 1	Primary analysis; column J
L2_SWI	0–100 (%); continuous	Percentage of 154 test words correctly transcribed by Listener 2	Primary analysis; column K

L3_SWI	0–100 (%); continuous	Percentage of 154 test words correctly transcribed by Listener 3	Primary analysis; column L. Listener 74's data for participant 30310 discarded due to audio playback error; participant 30310 contributes 2 listener observations rather than 3 in the LMM.
--------	--------------------------	--	---

EXPOSURES

Exposure (friendly name)	Values and Format	Definition	Notes or QC comments
Cohort	1=CLP, 2=CP, 3=TD, 4=OM; categorical	Child's diagnostic group: 1=repaired cleft lip and palate (CLP), 2=repaired cleft palate only (CP), 3=typically developing (TD), 4=early history of otitis media (OM)	Primary exposure; N=23, 24, 34, 34 respectively; column B
Cleft Type	1=none, 2=UCLP, 3=BCLP, 4=soft palate only, 5=soft & hard palate; categorical	1=none (TD and OM), 2=unilateral CLP, 3=bilateral CLP, 4=cleft soft palate only, 5=cleft soft and hard palate	CLP cohort = codes 2–3; CP cohort = codes 4–5; column C

COVARIATES

Covariates (friendly name)	Values and Format	Definition	Notes or QC comments
Sex	1=M, 2=F; categorical	Biological sex	column D
Age	Continuous; years	Age in years at time of recording	column E
VP Status	1=adequate, 2=inadequate; categorical	1=<5 mm ² (adequate VP closure), 2=>5 mm ² (inadequate VP closure)	6 CLP + 1 CP coded as inadequate; all TD and OM coded as adequate; column AC;

OTHER VARIABLES

Variable (friendly name)	Values and Format	Definition	Notes or QC comments
ID	Integer; cat- egorical	Unique child identifier; first two digits denote recruitment site	3 recruitment sites; column A
L1, L2, L3	Integer; cat- egorical	IDs of listeners 1–3 assigned by transcription program (not se- quential)	66 total listeners across 22 groups; columns F–H;
L_Group	1–22; cate- gorical	Listener group assignment (3 lis- teners per group, 5–6 children per group)	≥ 2 children with clefts per group; column I
L1.L2, L1.L3, L2.L3	0–100 (%); continuous	Absolute pairwise differences in SWI scores between each pair of listeners	Listener 74 data for participant 30310 discarded due to audio playback error in transcription program; columns N–P
L_Avg	0–100 (%); continuous	Mean of absolute pairwise lis- tener differences (L1.L2, L1.L3, L2.L3)	Inter-listener agreement check; values >10% flagged and high- lighted in Excel; column Q

PRELIMINARY ANALYSIS

Key decisions

1. **TD vs. OM comparability.** Prior to primary analysis, test whether the typically developing (TD, cohort 3) and otitis media (OM, cohort 4) groups differ on the primary outcome (M.SWI). Use an independent-samples *t*-test (or Wilcoxon rank-sum if normality is not met). If no statistically significant differences are found (and effect sizes are negligible), the two groups will be merged into a single control group for the primary analysis. Document this decision and retain the original cohort coding as a sensitivity check. In terms of whether this accounts for alpha-spending, TBD.
2. **Inter-rater agreement by listener group.** Assess agreement among the three listeners within each of the 22 listener groups using intraclass correlation coefficient (ICC, two-way mixed model, absolute agreement). Flag listener groups where CC falls below < 0.5 , indicating poor agreement (see REFERENCES). Results will inform:
 - Whether individual listener scores (L1.SWI, L2.SWI, L3.SWI) exhibit meaningful clustering within listener groups, motivating the use of a linear mixed model in the primary analysis.
 - Identification of listener groups with low agreement for the sensitivity analysis (see Sensitivity Analysis, item 2).

Note: Listener 74 data for participant 30310 have already been discarded due to an audio playback error in the transcription program.

PRIMARY ANALYSIS

Key decisions

Linear mixed model (LMM). Fit an LMM using *individual listener SWI scores* (L1.SWI, L2.SWI, L3.SWI; long format, one row per listener per child) as the outcome. Fixed effects: cohort, sex, age, VP status. We will also consider pairwise interactions between the fixed effects. Random effects: listener nested within listener group (i.e., random intercept for listener group, with listener as the unit of replication within group). Use REML estimation.

Missing Data

The full analytic sample of $N=115$ is retained; no participants are excluded. The only data anomaly is that responses from Listener 74 for participant 30310 were discarded prior to data receipt due to an audio playback error in the transcription program.

Significance level / Multiple Comparisons

Depending on the results of the preliminary analysis, we will conduct either two or three planned pairwise contrasts against the CLP group, extracted after fitting the LMM. If TD and OM are merged into a single control group, two contrasts will be tested (CLP vs. CP; CLP vs. control). If TD and OM are retained as separate cohorts, three contrasts will be tested (CLP vs. CP; CLP vs. TD; CLP vs. OM).

We will use the weighted Bonferroni-Holm procedure to adjust for multiplicity. The initial allocation of significance to the individual hypothesis is α/m , where m is the number of primary hypothesis. Suppose we have three hypothesis H_1, H_2, H_3 . If H_1 is rejected at α/m , then the associated significance is split equally and passed on to the remaining (not yet rejected) hypothesis H_2 and H_3 . This will continue until all the remaining α is allocated.

Multiplicity is controlled separately within the primary and secondary analyses rather than jointly across both. This decision is justified by the conceptual distinction between the two sets of hypotheses: the primary analysis targets overall speech intelligibility (individual listener scores), while the secondary analysis targets contrast-specific intelligibility scores (alv_pal through vd_vl), which represent distinct research questions.

Sensitivity Analysis

1. **Simpler random effects.** Compare the primary LMM with a LMM with random effects only for the listener group. Use likelihood ratio test if the difference in the log-likelihood and use a 50-50 mixture of a χ^2_1 and a χ^2_2 distribution to obtain the critical value.
2. **Exclude participants with inadequate VP closure.** Rerun the LMM model excluding the 7 participants with VP status = 2 (inadequate; $> 5\text{mm}^2$): 6 from the CLP cohort and 1 from the CP cohort. Compare estimates and standard errors to the main analysis to assess the influence of velopharyngeal inadequacy on the cohort effect.
3. **Ordinary linear regression (OLS).** Regress M_SWI (mean intelligibility score across three listeners) on cohort (reference = control group, per preliminary analysis decision above), sex, age, and VP status. Assess potential two-way interactions (e.g., cohort $\times$ age, cohort $\times$ VP status) based on preliminary data inspection.

SECONDARY ANALYSIS

Key decisions

Contrast-specific intelligibility by cohort. For each of the 11 phonetic contrast scores listed below, fit a separate OLS linear regression with the contrast score as the outcome and the same fixed effects (and interaction terms) as the primary model (cohort, sex, age, VP status). The number of cohorts (3 or 4) follows the decision made in the preliminary analysis regarding merging of the TD and OM groups.

Variable	Contrast
alv_pal	Alveolar-palatal sibilant (e.g., <i>see-she</i>)
alv_velar	Alveolar-velar stop (e.g., <i>tap-cap</i>)
cl_sing	Cluster-singleton (e.g., <i>stop-top</i>)
con_null	Initial consonant-null (e.g., <i>pat-at</i>)
fr_bk	Front-back vowel (e.g., <i>fill-full</i>)
fric_affr	Fricative-affricate (e.g., <i>ship-chip</i>)
hi_lo	High-low vowel (e.g., <i>beat-bit</i>)
r_w	Liquid-glide (e.g., <i>ray-way</i>)
st_fric-affr	Stop-fricative/affricate (e.g., <i>base-vase</i>)
st_nas	Stop-nasal (e.g., <i>bop-mop</i>)
vd_vl	Voiced-voiceless stop (e.g., <i>big-pig</i>)

Note: Each contrast score is already averaged across the three listeners (analogous to M_SWI), so the OLS model is appropriate.

Significance level / Multiple Comparisons

As stated above, no gatekeeping approach is applied between the primary and secondary analyses. Within the secondary analysis, however, we apply a gatekeeping structure across two families of hypotheses.

Family 1 (CLP vs. control): For each of the 11 phonetic contrasts, test whether the CLP cohort differs from the control group. The 11 p -values are adjusted using the Hochberg step-up procedure at two-sided $\alpha = 0.05$. The Hochberg procedure is appropriate because children who score highly on one contrast are expected to score highly on others, inducing positive dependence among the test statistics.

Family 2 (CLP vs. CP): Only contrasts for which Family 1 was rejected proceed to Family 2 testing. For those contrasts, we test whether the CLP cohort additionally differs from the CP cohort, again using the Hochberg step-up procedure at $\alpha = 0.05$ applied to the reduced set of hypotheses. Contrasts not rejected in Family 1 are not tested in Family 2.

This gatekeeping structure reflects the natural scientific hierarchy of the research questions: we first identify which phonetic contrasts distinguish CLP from typical speech, then ask whether those same contrasts also differentiate CLP from another clinical group (CP). Note that this approach cannot detect contrasts where CLP differs from CP but not from the control group.